

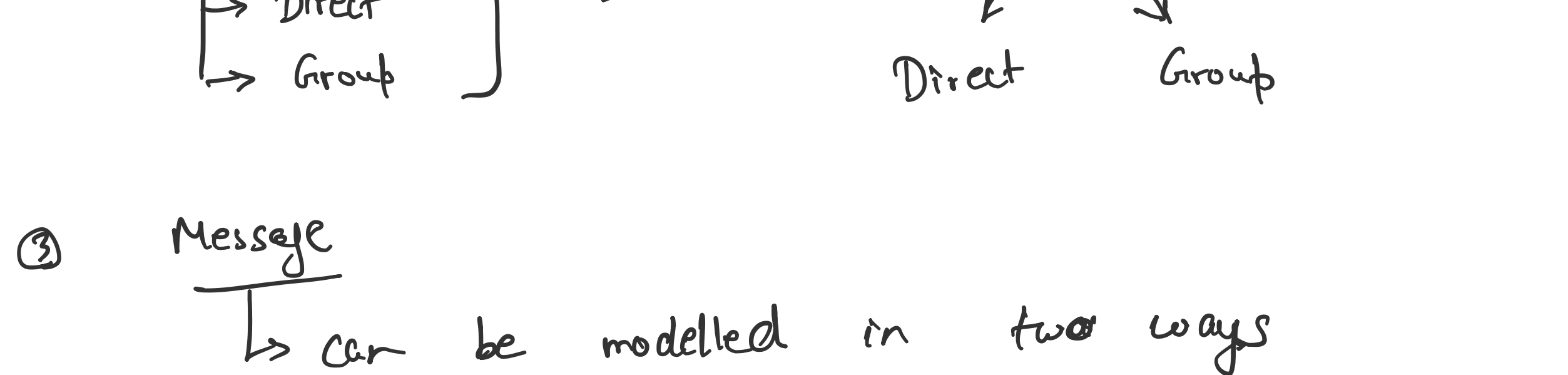
Requirements

- 1-1 messaging
- Group messaging
- Message delivery status (Message Receipt / Read Receipt)
- Media messages (Audio / Video / Image / File)
- User status (Online / Offline)
- Notifications

Out of scope

- ✓ Voice calls
- ✓ Disappearing messages
- ✓ Encryption
- ✓ Status

Core Entities



③ Message

↳ can be modelled in two ways

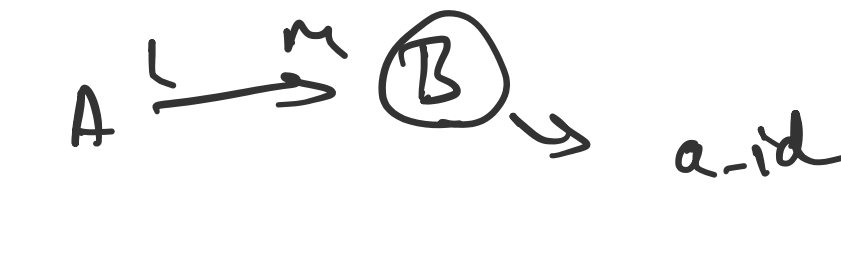


④ Notification



Table

↳ keep id on many side



Chat

name

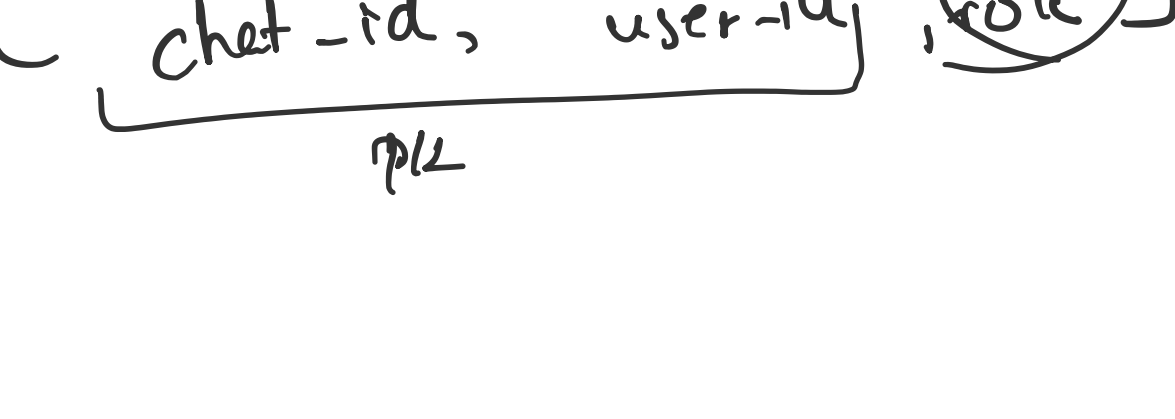
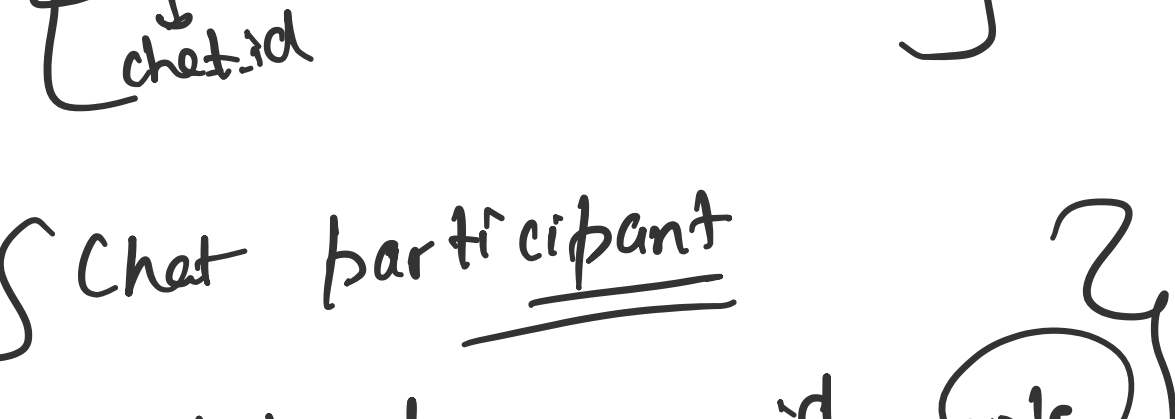
chat-id

type

List < User > participants;

(Map < User, role > participants)

List < ChatParticipants >



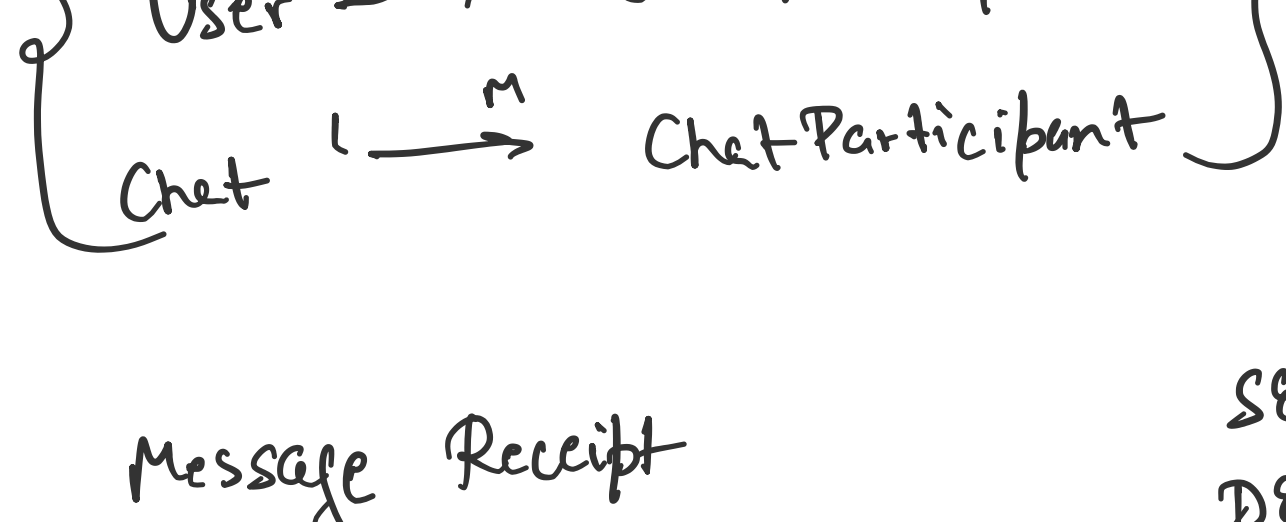
Chat

↳ admin

Chat $\rightarrow$ participants $\rightarrow$ User

Chat $\rightarrow$ ChatParticipants $\rightarrow$ filter by Role $\rightarrow$ User

admins

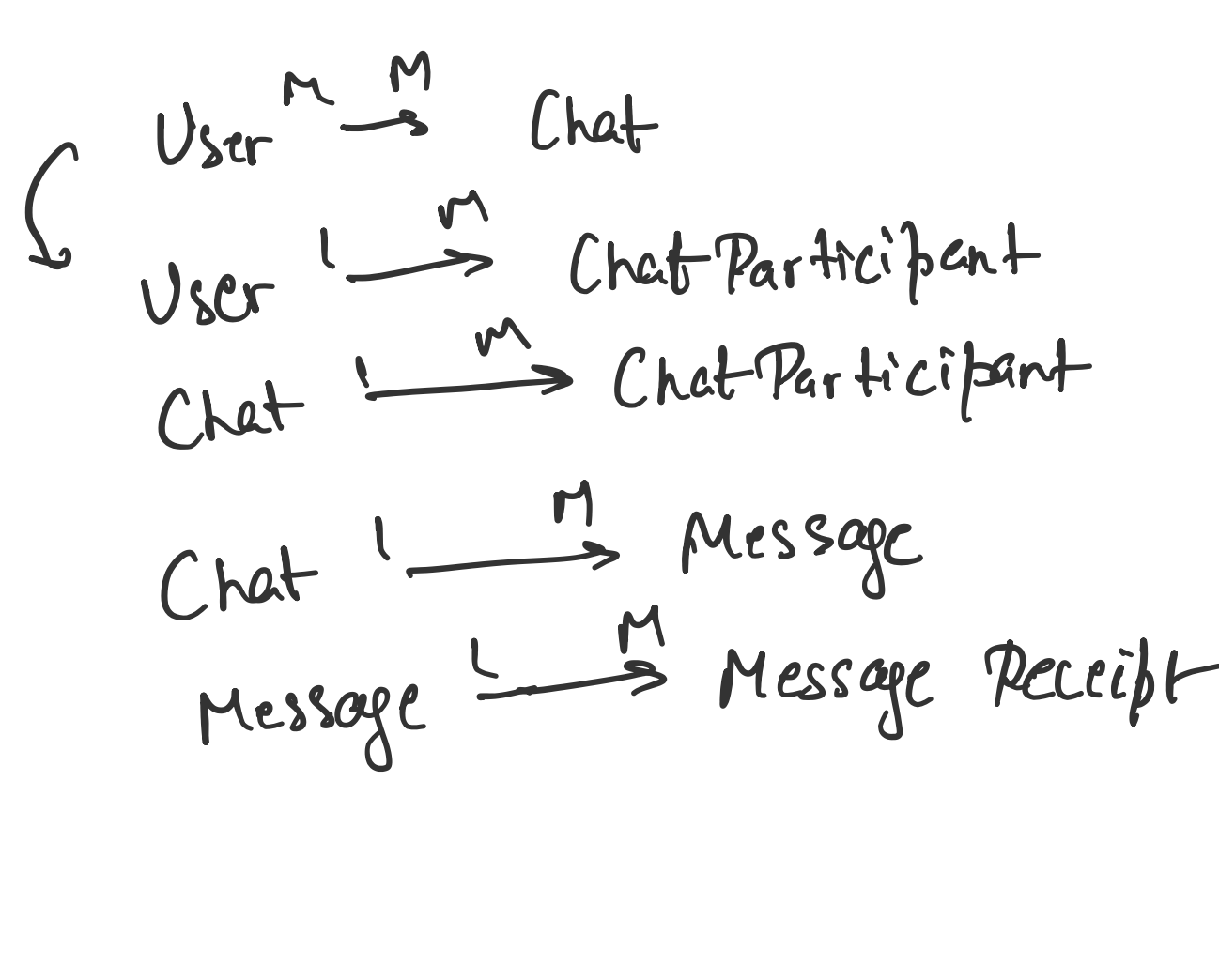
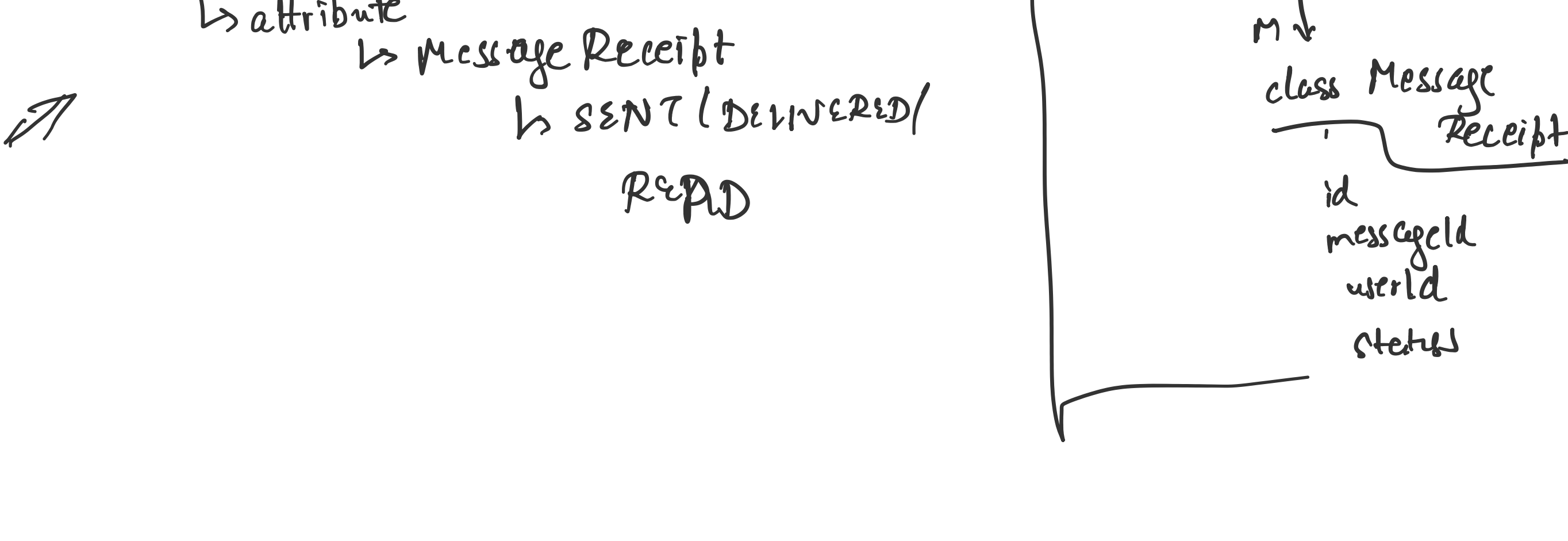


Message Receipt

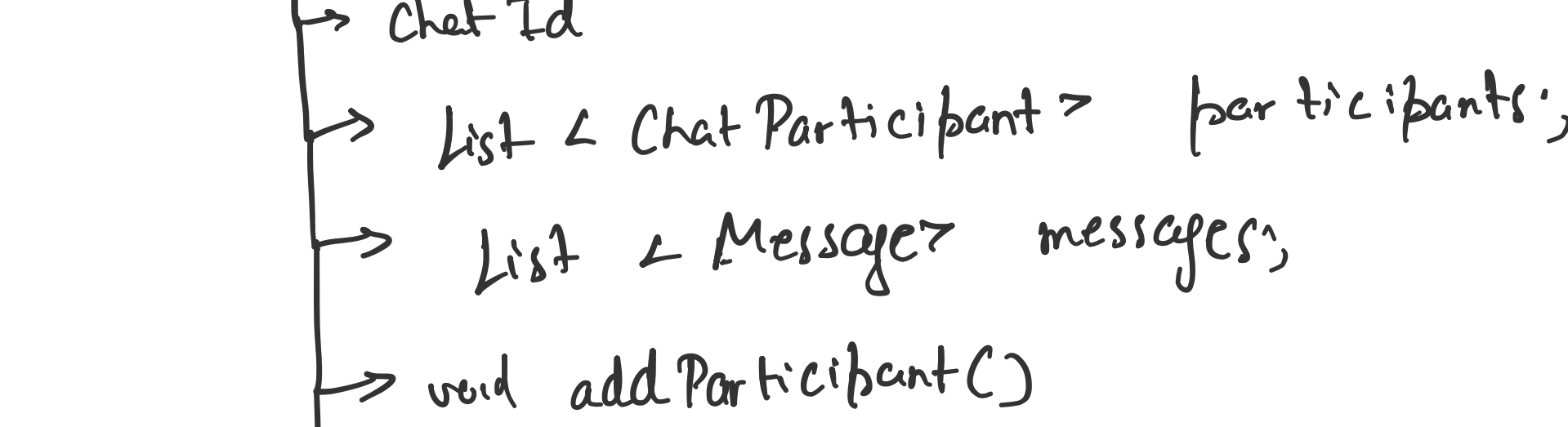
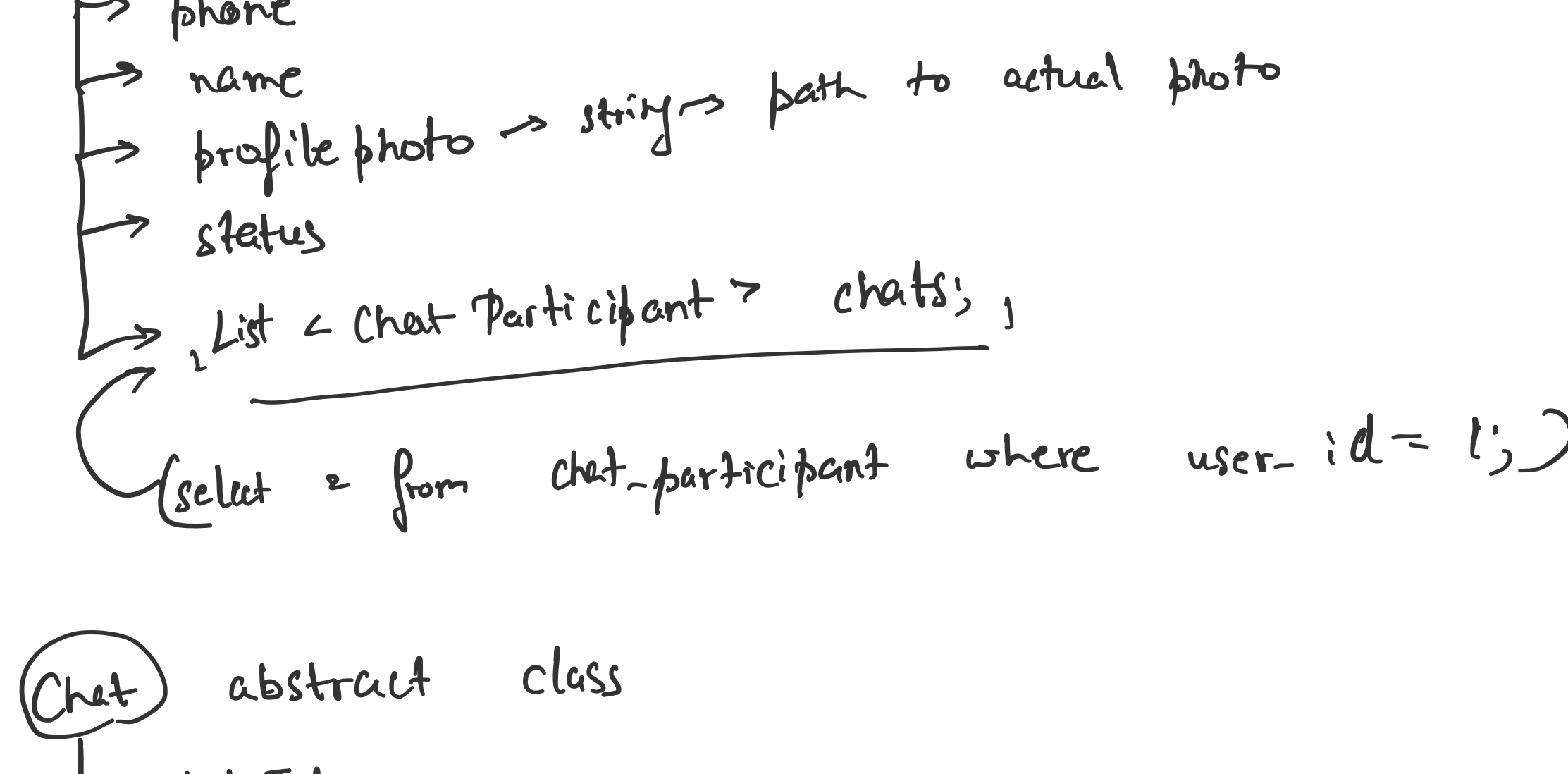
SENT

DELIVERED

READ



Classes



② Direct chat (extends) Chat

User user

User user2

optional

③ Group Chat

groupname

add Admin (User) $\rightarrow$ optional

④ Chat Participant

id

User

Chat

Participant Role role

Joined At

left At

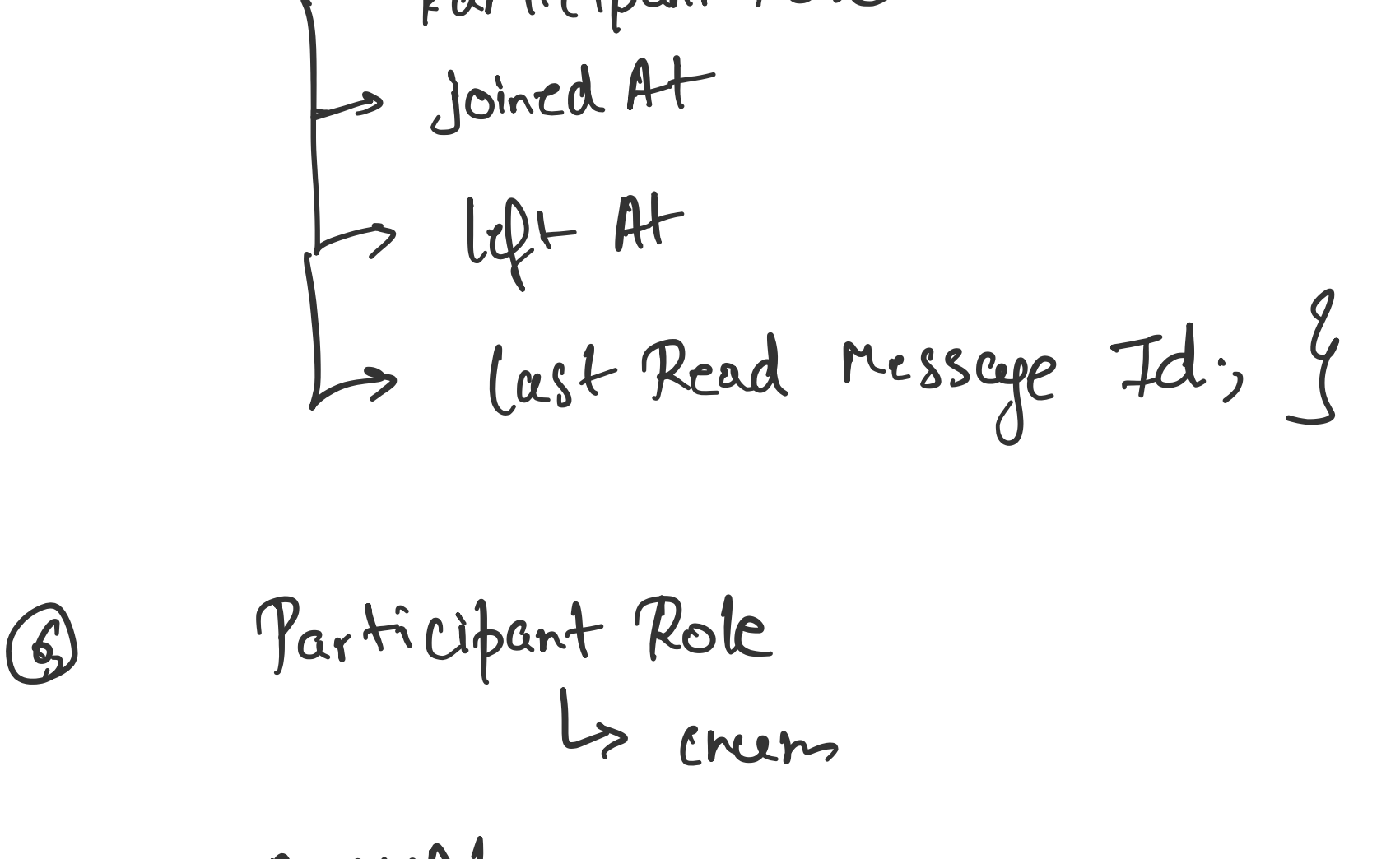
Last Read Message Id;

⑤ Participant Role

enum

ADMIN

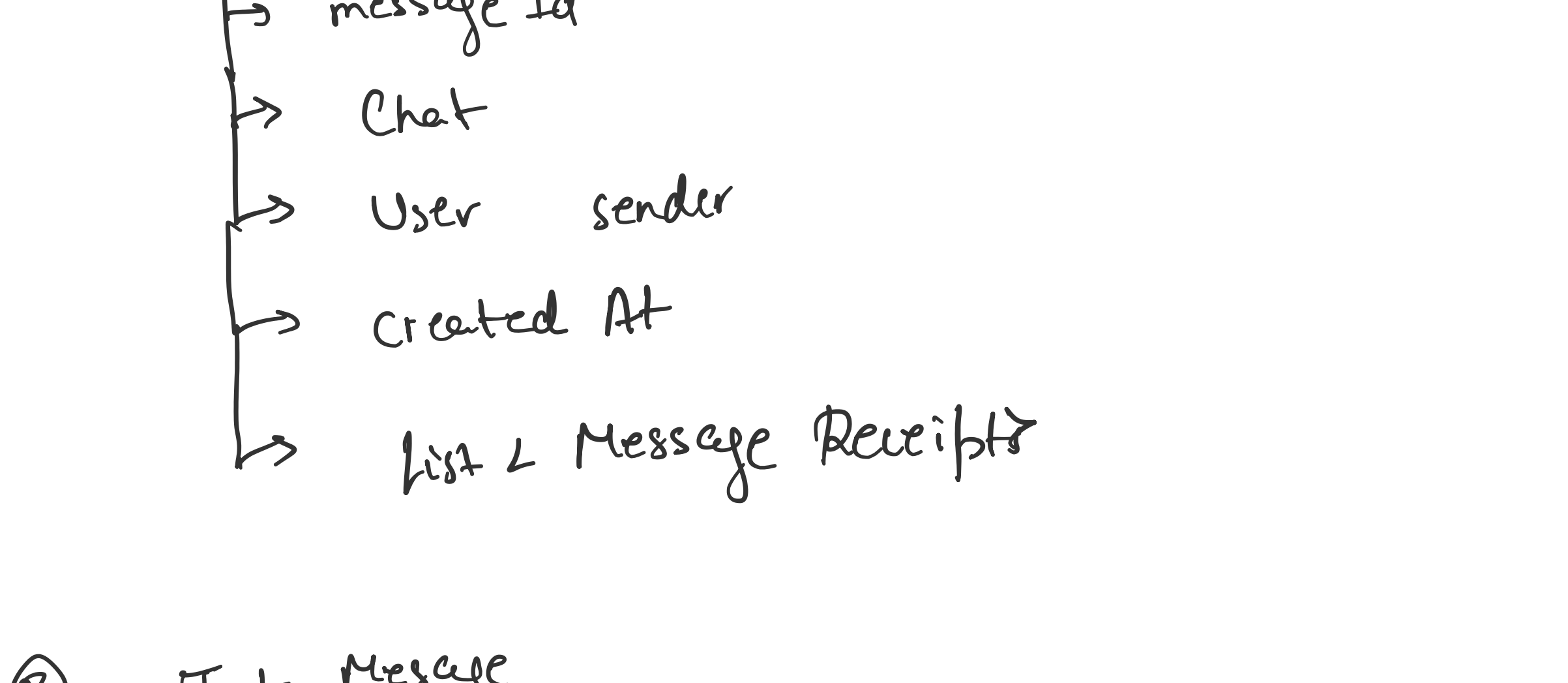
MEMBER



⑦ Text Message

⑧ Audio Message

⑨ Media Message



⑪ Receipt Status

enum

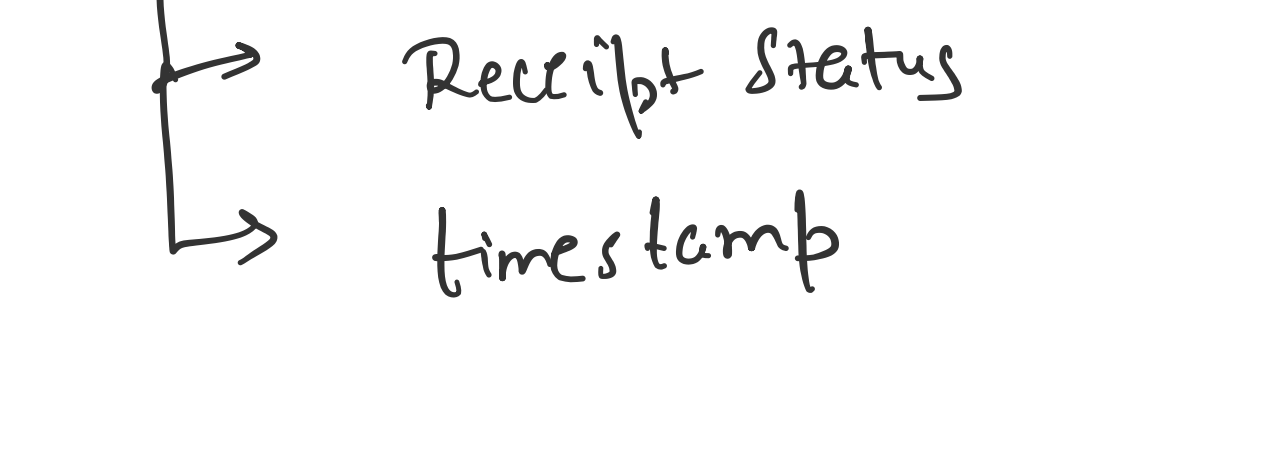
SENT

DELIVERED

READ

Services

Chat Service



Messaging Service

$\rightarrow$ send Message (chat, sender, message)

Receipt Service



Notification Service

$\rightarrow$ notify Users (List < User > message)

Message Factory

creation of individual messages.

